Theorem 2. *Suppose that the initial data are generated independently according to the model $y \mid x \sim \mathsf{N}(x^\top \theta, \sigma^2 \mathrm{I}_d)$ where $\theta \in \mathbb{R}^d$ is a fixed parameter. If $\boldsymbol{G}_1, \ldots, \boldsymbol{G}_t$ are full rank then*

$$\mathbb{E}[\hat{\boldsymbol{\theta}}_t] = \big(\mathrm{I} - \boldsymbol{Q}_t \cdots \boldsymbol{Q}_1(\mathrm{I} - \boldsymbol{G}_0 \boldsymbol{G}_0^+)\big)(\mathbf{1}_K \otimes \theta), \quad \mathsf{Cov}(\hat{\boldsymbol{\theta}}_t) = \boldsymbol{M}_t \begin{bmatrix} \tilde{\boldsymbol{S}}^+ & 0 \\ 0 & S_*^+ \end{bmatrix} \boldsymbol{M}_t^\top + \boldsymbol{C}_t$$

To help interpret this result, observe that if $\boldsymbol{G}_0$ is full rank, then each initial estimate is unbiased, and unbiasedness persists throughout every stage of training. Conversely, if $\boldsymbol{G}_0$ is rank deficient, then at least one (and possibly all) of the initial estimates is biased. Remarkably, Theorem 2 shows that it may still be possible for all entities have vanishing bias, provided that $\boldsymbol{Q}_t \cdots \boldsymbol{Q}_s$ converges to zero. Specific conditions under which this occurs are considered in the next section.

Asymptotic Variance. To provide a finer analysis of the training dynamics we now suppose that the weights and features satisfy $\alpha_t = \alpha$, $\beta_t = \beta$, and $\boldsymbol{S}_t = \boldsymbol{S}$ for $t \geq 1$. Setting $\boldsymbol{P} = \boldsymbol{P}_1$ and $\boldsymbol{Q} = \boldsymbol{Q}_1$, the matrices $\boldsymbol{M}_t$ and $\boldsymbol{C}_t$ defined in Theorem 1 can be expressed explicitly as

$$\boldsymbol{M}_t = \boldsymbol{Q}^t \boldsymbol{P}_0 + \Big(\sum_{s=0}^{t-1} \boldsymbol{Q}^s\Big)\boldsymbol{P}, \qquad \boldsymbol{C}_t = \sigma^2 \sum_{s=1}^{t} \boldsymbol{Q}^s \boldsymbol{S}^+ \big(\boldsymbol{Q}^s\big)^\top \tag{1}$$

Classical results in matrix analysis (Higham, 2002) imply that if the spectral radius of $\boldsymbol{Q}$ is strictly less than one, then these these matrices converge to well-defined limits $\boldsymbol{M}$ and $\boldsymbol{C}$ satisfying

$$\boldsymbol{M} \coloneqq (\mathrm{I} - \boldsymbol{Q})^{-1}\boldsymbol{P}, \qquad \mathsf{vec}(\boldsymbol{C}) \coloneqq \sigma^2 (\mathrm{I} - \boldsymbol{Q} \otimes \boldsymbol{Q})^{-1}\, \mathsf{vec}(\boldsymbol{Q}\boldsymbol{S}^+\boldsymbol{Q}) \tag{2}$$

The following result provides a sufficient condition for convergence in terms of the triple $(\tilde{\boldsymbol{S}}, S_*, \boldsymbol{S})$. In particular, if $\boldsymbol{S}$ is proportional to $\tilde{\boldsymbol{S}}$ then the condition is satisfied for all $0 \leq \alpha < 1$ and $0 \leq \beta \leq 1$. Note that the boundary case $\alpha = 1$ corresponds to the recursive training setting of Shumailov et al. (2024) where the variance increases linearly across generations, and thus convergence does not occur.

Lemma 1. *Suppose that $\boldsymbol{S} \propto \lambda \tilde{\boldsymbol{S}} + (1-\lambda)(\mathrm{I}_K \otimes S_*)$ for some $0 < \lambda \leq 1$. Then, the spectral radius of $\boldsymbol{Q}$ is strictly less than one for all $0 \leq \alpha < 1$ and $0 < \beta \leq \lambda$.*

We summarize our findings with the following characterization of the asymptotic variance:

Theorem 3. *Consider the setting of Theorem 2 and suppose that $\alpha_t = \alpha$, $\beta_t = \beta$, and $\boldsymbol{S}_t = \boldsymbol{S}$ for $t \geq 1$. If $\boldsymbol{G} = \boldsymbol{G}_1$ has full rank and $\boldsymbol{Q}$ has spectral radius strictly less than one, then*

$$\mathbb{E}[\boldsymbol{\theta}_t] \xrightarrow{t\to\infty} \mathbf{1}_K \otimes \theta, \qquad \mathsf{Cov}(\hat{\boldsymbol{\theta}}_t) \xrightarrow{t\to\infty} \sigma^2 \boldsymbol{M} \begin{bmatrix} \tilde{\boldsymbol{S}}^+ & 0 \\ 0 & S_*^+ \end{bmatrix} \boldsymbol{M}^\top + \boldsymbol{C}$$

where $\boldsymbol{M}$ and $\boldsymbol{C}$ are given by (2).

MSE and relative efficiency. The expression for the mean and variance in Theorems 2 and 3 provide explicit formulas for the mean squared error (MSE) $\mathbb{E}[\|\hat{\theta}_{tk} - \theta\|^2]$ of entity k at each generation t and the mean squared prediction error (MSPE) $\mathbb{E}[\|\tilde{X}_m(\hat{\theta}_{tk} - \theta)\|^2]$ for entity m's private feature matrix.

We can use this to compute the optimal α, β values for the interactive training setting. Figure 2 compares the asymptotic MSE for a training workflow with given α, β (from Theorem 3) with the MSE of an idealized setting where each entity has access to the entire collection of real data (both private and public). Each curve represents the relative efficiency, i.e., the ratio of optimal MSE to entity-specific workflow MSE, with values close to one indicating near optimality. These results demonstrate that a setting with $\beta = 0.5$ and $\alpha = 0.5$ achieves the best global performance for all models. When β is much larger (0.9), relatively small α values also improve model performance.

6 Experimental Evaluation

To understand how our theoretical predictions bear out in practice, we run experiments on text-generation models. In each, we train K interacting models (per our framework in Figure 1) for several generations and evaluate how their performance changes on their own and other models' tasks. Here, we present the setup and results for a $K = 2$ model system. Additional results are in the Appendix.

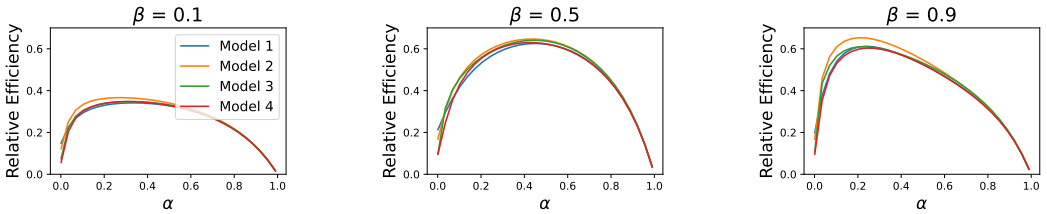

Figure 2: **Predicted relative efficiency across α, β values for a $K = 4$ model system (dimension 15 and rank 5).** *Curves show ratio of MSE of the minimum variance unbiased estimator to asymptotic MSE for a given α, β setting obtained from Theorem 3 (1 is optimal). Setting $\alpha = \beta = 0.5$ produces best results across models.*

Experiment setup. Training large language models from scratch is computationally infeasible for us, so we simulate the initial setup of language models trained on dataset $(D_*, \tilde{D}_k)$ at time $t = 0$ by fine-tuning $K = 2$ instances of a given pre-trained model architecture on carefully chosen $D_*, \tilde{D}_k$. We experiment with two language model architectures—OPT-350m (Zhang et al., 2022) and Llama 3.2 1B (Dubey et al., 2024)—to assess how interactive training scales. Public sources (Touvron et al., 2023a; Zhang et al., 2022) state that both models were pretrained on BookCorpus (Yang, 2022), CC-Stories (Anderson, 2022), the English portion of CommonCrawl, and public Reddit data. We approximate D_* with BookCorpus due to practical constraints. Each model is given its own initial task-specific dataset $\tilde{D}_k$ —SciQ (Johannes Welbl, 2017) (science questions) for the $k = 0$ model; and OpenAI's GSM8K (Cobbe et al., 2021) (grade school level math problems) for the $k = 1$ model.

Interactive training proceeds as outlined in Figure 1, with fixed α, β values for each experiment. After training a new model generation $\hat{\theta}_{tk}$, we use $\hat{\theta}_{tk}$ to produce synthetic data $D_{t+1,k}$ that becomes part of the next generation's training data (if $\alpha > 0$). We produce $D_{t+1,k}$ by randomly sampling prompts from $\tilde{D}_k$ for each of the K models and prompting $\hat{\theta}_{tk}$ to complete the text.

Training and evaluation. We run experiments on $K = 2$ model systems with $\alpha \in \{0, 0.5, 1\}$ and $\beta \in \{0, 0.5, 1\}$, each for $T = 15$ generations of training. At each training generation, models are fine-tuned on datasets of fixed size $n = 12,500$ drawn i.i.d. from the datasets $\tilde{D}_k$, D_*, D_t with weights $\bar{\alpha}\beta$, $\bar{\alpha}\bar{\beta}$, and α/K, respectively. This effectively mimics the *accumulate and subsample* setup of Kazdan et al. (2025) with the additional wrinkle of data-mediated model interactions.

We train each model for 100 steps per generation on a single NVIDIA H200 GPU using mixed-precision, the AdamW optimizer with a learning rate of $8e^{-6}$, warmup ratio of 0.025, and gradient accumulation over 2 steps. After training each generation, we record evaluation loss (token-wise average cross-entropy loss) by feeding each model prompts from each test set of private data $\tilde{D}_k$ and evaluating semantic "distance" between predicted and correct answer. We also compute embedded representations of models' completions of the first 200 elements of each of $\tilde{D}_k$ and D_*, so we can measure how various α, β affect models' representational spaces. Embeddings are computed via the `SentenceTransformers` python library. If models produce outputs with similar embeddings (as measured by cosine similarity), their feature spaces are more aligned. These two metrics allow us to evaluate how data-mediated interactions between models affect (1) models' performance on their own and other models' tasks and (2) model homogeneity as measured by embedding closeness.

Table 3: **Change in loss behavior for $K = 2$ interacting models at $\beta = 0.5$.** *We show results as `initial` → `final` prediction loss values for models on their own and the other models' tasks, at `initial` $= T = 0$ and `final` $= T = 15$ generations. For clarity, we colorize loss increase, decrease, and constancy ($\Delta \leq 0.1$).*

(a) *OPT models*

	$\alpha = 0$	$\alpha = 0.5$	$\alpha = 1.0$
Model 1 on Task 1	3.3 → 3.3	3.3 → 3.3	3.3 → 3.5
Model 2 on Task 2	1.8 → 1.7	1.8 → 1.7	1.8 → 2.2
Model 1 on Task 2	3.1 → 3.5	3.1 → 1.8	3.1 → 2.2
Model 2 on Task 1	5.1 → 5.1	5.1 → 3.5	5.1 → 3.5

(b) *Llama 3.1 models*

	$\alpha = 0$	$\alpha = 0.5$	$\alpha = 1.0$
Model 1 on Task 1	2.8 → 2.9	2.8 → 2.9	2.8 → 3.0
Model 2 on Task 2	1.2 → 1.2	1.2 → 1.3	1.2 → 1.9
Model 1 on Task 2	2.0 → 2.5	2.0 → 1.4	2.0 → 1.8
Model 2 on Task 1	3.7 → 4.2	3.7 → 3.0	3.7 → 3.0

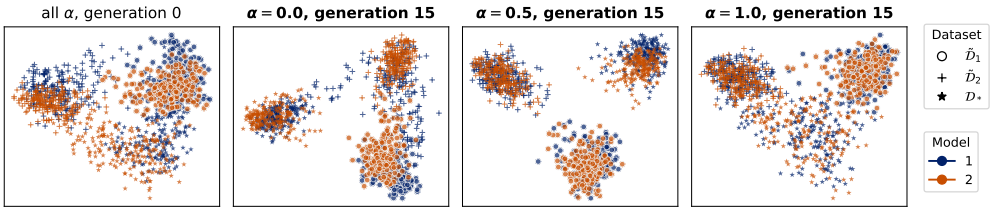

Figure 3: **PCA of embeddings of outputs produced by models θ_{t1} and θ_{t2} on datasets $\tilde{\mathcal{D}}_1$, $\tilde{\mathcal{D}}_2$, and $\tilde{\mathcal{D}}_*$. Results for $K = 2$ Llama models trained with $\beta = 0.5$ and varying α.** *Leftmost plot shows embeddings at generation $t = 0$, which are identical for all α; while right plots show embeddings at $t = 15$ with different α.*

Table 4: **Cosine similarity of embedded output representations for $K = 2$ Llama models at $\beta = 0.5$.** *We show* `initial(t=0)` $\rightarrow$ `final(t=15)` *for $\tilde{\mathcal{D}}_1$ $\tilde{\mathcal{D}}_2$, and $\mathcal{D}_*$. We colorize increase and decrease.*

	$\alpha = 0$	$\alpha = 0.5$	$\alpha = 1.0$
$\tilde{\mathcal{D}}_1$	$0.73 \rightarrow 0.70$	$0.73 \rightarrow 0.88$	$0.73 \rightarrow 0.86$
$\tilde{\mathcal{D}}_2$	$0.88 \rightarrow 0.79$	$0.88 \rightarrow 0.93$	$0.88 \rightarrow 0.94$
$\mathcal{D}_*$	$0.50 \rightarrow 0.59$	$0.50 \rightarrow 0.64$	$0.50 \rightarrow 0.58$

Results. As predicted in Figure 2, an $\alpha = 0.5$, $\beta = 0.5$ setting produces optimal results in terms of model performance across tasks. Table 3 reports the change in models' loss values between the first and last ($t = 15$) training generation for the $\beta = 0.5$ setting. When $\alpha = 0$, the same (human-generated) data is used for each training update, resulting in stable or slightly worse performance on different tasks, as models are forced into a local minima. When $\alpha = 1.0$, models are only trained on generated outputs. They degrade on their original task due to the lack of real data but improve slightly on the other model's task. However, at $\alpha = 0.5$, models perform well on their own tasks and improve on the other model's task. Results for other β and $K = 3$ are in Appendix and echo findings here.

While some amount of mixing improves model performance on previously-unseen tasks, homogenization occurs for D_* at all α and for $\tilde{D}_k$ when $\alpha > 0$—everywhere it can. Figure 3 shows the benefit of interactive training when α, β are reasonable, visualizing PCA-reduced embeddings from $\tilde{D}_1$, $\tilde{D}_2$, and D_* for $\beta = 0.5$ and varying α as training progresses For $\alpha = 0.5$, the PCA shows clearly separated task clusters, indicating that both models are better tailored to the individual tasks. In contrast, the lack of well-defined clusters when $\alpha \neq 0.5$ suggests reduced sensitivity to task type, e.g. generic answers. Yet, in Table 4, we see that when $\alpha = 0$, models homogenize slightly on the shared task D_* but diverge on model-specific tasks. This makes sense since models the models do not train on each other's tasks. Once $\alpha > 0$, homogenization increases for all datasets/tasks.

7 Discussion

Limitations. Our work has several limitations. We discuss possible objections to our claims about the increasing presence of generated outputs in training datasets (e.g. (Drayson et al., 2025)) in Appendix B. Also, our theoretical framework assumes that new data in model updates is purely synthetic. In reality, if internet scrapes are used to create model update datasets, they will contain both synthetic and real data. Finally, we assume that model trainers use new scraped data for each model update but only reuse data from initial training. This assumption may limit the range of outcomes.

Broader Impacts. If data-mediated interactions homogenize generative models, causing them to coalesce on certain viewpoints, this could lead to pervasive bias in AI-generated content. Peterson (2025) discusses this possibility, while Wenger & Kenett (2025) showed homogeneity across creative outputs from many LLMs, suggesting these homogenization effects may already be felt by models. Much future study is needed to evaluate the extent to which data-mediated interactions fuel homogeneity (as opposed to other causes) and develop mitigations.

Conclusions and Future Work. We provide a first look at possible outcomes of genAI models trained on each others' data and find mixed effects. Training on other models' data exposes models to concepts possibly missed in their own training data, but can homogenize model behaviors. Future work could consider additional nuances of interactions between models, explore how these interactions evolve in other modalities like image generation, and investigate whether fixed points (e.g. like the universal $\pi^2/6$ pathway of (Dey & Donoho, 2024)) exist under this paradigm.